USRP Studies

Week 4: Spectral Graph Theory Continued (part b)

Overview

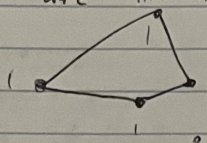
I continued the lecture about spectral graph theory. This part was probably the most difficult topic to grasp for me, because it was much more abstract than the other lectures. In fact, I didn't fully understand some proofs and relied on 'heuristics' for a few parts. There were also a few linear algebra concepts I had to brush up on. The graph theory part was quite intuitive. Furthermore, although this is an interesting topic that I would love to learn more about, I think the payoff compared to opportunity cost of continuing is too low. I have the very basic idea and motivation for spectral graph theory which I think is enough, so I decided to move on to other topics.

Spectral Graph Thory (Lec 13) part 6

Recall: $\mathcal{E}[f] = \frac{1}{2} \mathop{E}_{uv}[(f(u)-f(v))^2]$,

Which f min/max. $\mathcal{E}[f]$?

Min: If conn. components of G are $S_1, \dots, S_k$ then $1_{S_1}, \dots, 1_{S_k}$ are lin. indep. and span f s.t. $\mathcal{E}[f] = 0$.



$\mathcal{E}[f] = 0$ iff f is const. on each conn. comp.

↙ squared norm

Max: Constrained w. $\|f\|_2^2 = \mathop{E}_{uv}[f(u)^2] \le 1$.

We proved that $\mathcal{E}[f] \le 2\|f\|_2^2$, w. eq. iff G is bipartite.

defn Let $f: V \to \mathbb{R}$. $Kf: V \to \mathbb{R}$ s.t. $(Kf)(u) = \mathop{E}_{uv}[f(v)]$.

K: normalized adjacency operator

"K basically replaces each vertex's value with the avg of its neighbors values".

other terms: Markov operator, transition operator.

To satisfy $Kf(u) = \mathop{E}_{uv}[f(v)]$,

let $K_{uv} = \frac{\#\text{edges between } u,v}{\deg(u)}$

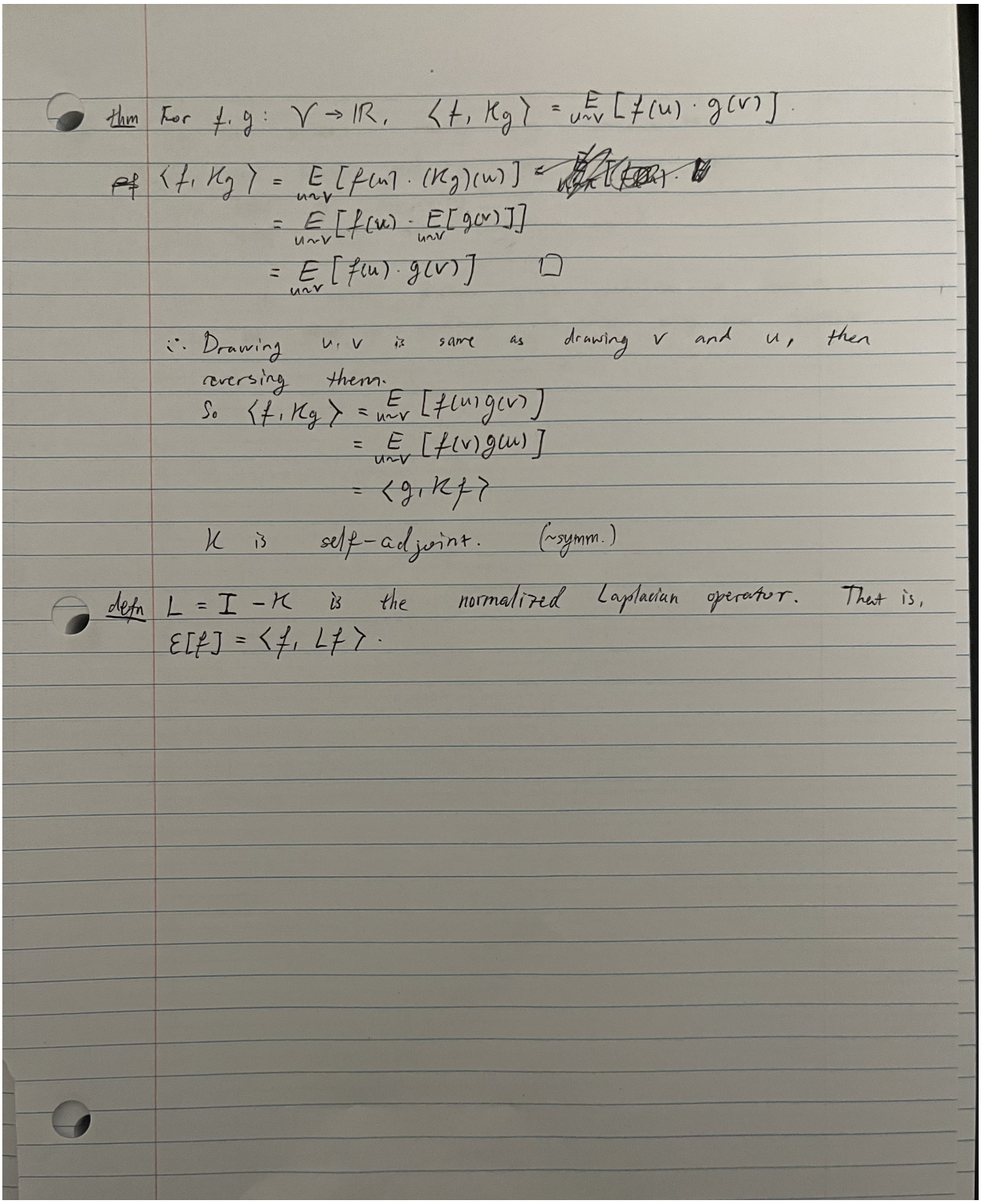
$\therefore$ G is d-regular $\Rightarrow$ K is the symmetric matrix $\frac{1}{d} A_G$

rmk K is linear.

$$[K][f] = [Kf].$$

defn G is <u>d-regular</u> when every vertex has same deg d.

defn A is <u>symmetric</u> when $A^T = A$.



thm For $f, g: V \to \mathbb{R}$, $\langle f, Kg \rangle = \mathop{E}_{u \sim v}[f(u) \cdot g(v)]$.

pf $\langle f, Kg \rangle = \mathop{E}_{u \sim v}[f(u) \cdot (Kg)(u)]$

$= \mathop{E}_{u \sim v}[f(u) \cdot \mathop{E}_{u \sim v}[g(v)]]$

$= \mathop{E}_{u \sim v}[f(u) \cdot g(v)]$ □

$\therefore$ Drawing u, v is same as drawing v and u, then reversing them.

So $\langle f, Kg \rangle = \mathop{E}_{u \sim v}[f(u) g(v)]$

$= \mathop{E}_{u \sim v}[f(v) g(u)]$

$= \langle g, Kf \rangle$

K is self-adjoint. (~symm.)

defn $L = I - K$ is the normalized Laplacian operator. That is, $\mathcal{E}[f] = \langle f, Lf \rangle$.